

EXP 3 TEST CASE 1

**Analysis of SWR (Standing Wave Ratio) pattern in High Frequency Transmission Lines
by varying a) load impedance b) Characteristic impedance**

PROGRAM :

```
clc;
clear;
close;

// Input Values
Zo = 50;    // Characteristic Impedance (Ohms)
ZL = 25;    // Load Impedance (Ohms)

// Reflection Coefficient

$$K = (ZL - Zo) / (ZL + Zo);$$


// Magnitude of Reflection Coefficient
ampK = abs(K);

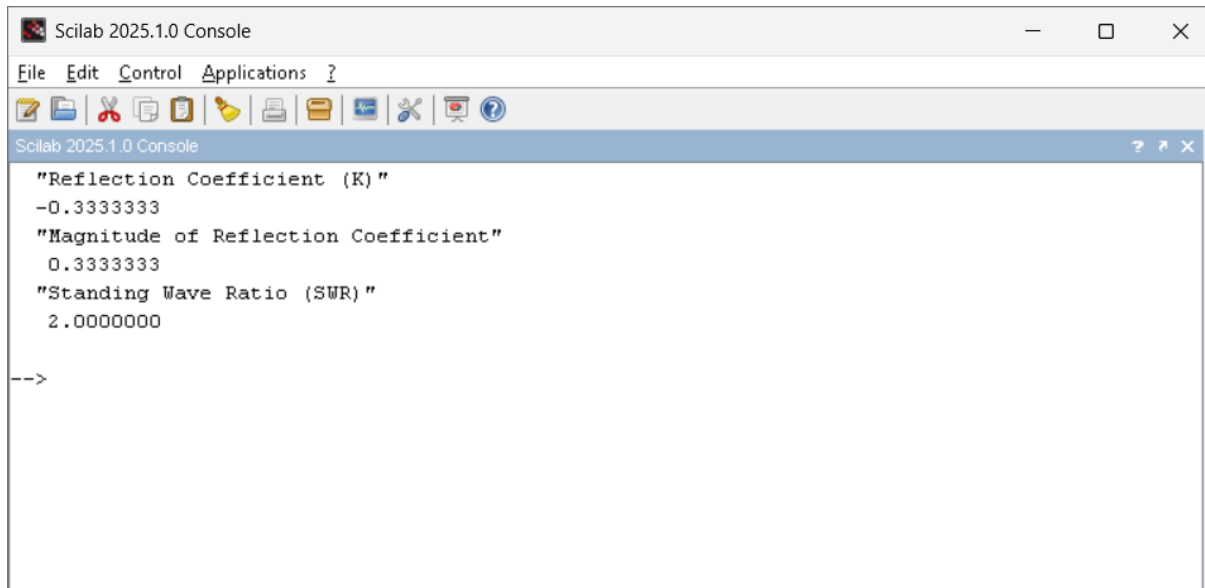
// Standing Wave Ratio (SWR)

$$SWR = (1 + ampK) / (1 - ampK);$$


// Output
disp("Reflection Coefficient (K)");
disp(K);

disp("Magnitude of Reflection Coefficient");
disp(ampK);

disp("Standing Wave Ratio (SWR)");
disp(SWR);
```

A screenshot of the Scilab 2025.1.0 Console window. The window has a title bar with the Scilab logo and the text "Scilab 2025.1.0 Console". Below the title bar is a menu bar with "File", "Edit", "Control", "Applications", and "?". Underneath the menu bar is a toolbar with various icons for file operations, editing, and help. The main area of the window is a text editor showing the output of a script. The output consists of four lines: "Reflection Coefficient (K) ", "-0.3333333", "Magnitude of Reflection Coefficient", and "0.3333333". Below these, it shows "Standing Wave Ratio (SWR) " and "2.0000000". At the bottom of the text area, there is a prompt "-->".

```
Scilab 2025.1.0 Console
File Edit Control Applications ?
"Reflection Coefficient (K) "
-0.3333333
"Magnitude of Reflection Coefficient"
0.3333333
"Standing Wave Ratio (SWR) "
2.0000000
-->
```

TEST CASE 2

`clc;`

`clear;`

`close;`

`// Input Values`

`ZL = 50; // Load Impedance (Ohms)`

`Zo = 250; // Characteristic Impedance (Ohms)`

`// Reflection Coefficient`

`K = (ZL - Zo) / (ZL + Zo);`

`// Magnitude of Reflection Coefficient`

`ampK = abs(K);`

`// Standing Wave Ratio (SWR)`

`SWR = (1 + ampK) / (1 - ampK);`

```
// Output
```

```
disp("Reflection Coefficient (K)");
```

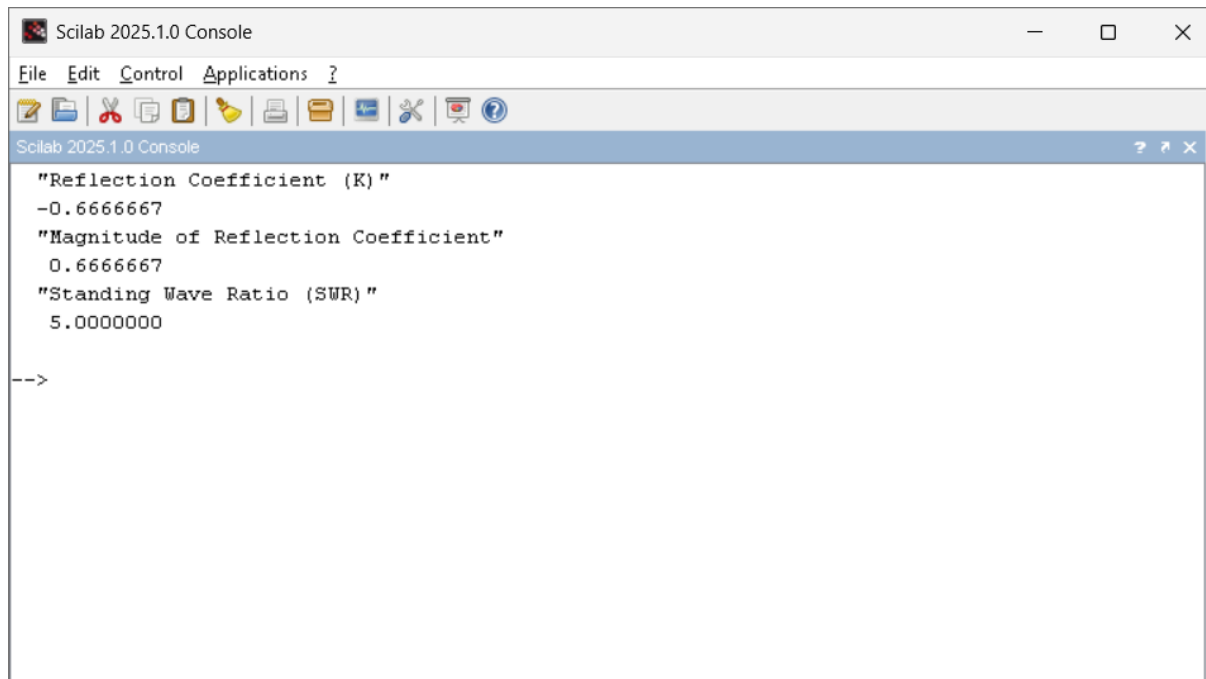
```
disp(K);
```

```
disp("Magnitude of Reflection Coefficient");
```

```
disp(ampK);
```

```
disp("Standing Wave Ratio (SWR)");
```

```
disp(SWR);
```



The image shows a screenshot of the Scilab 2025.1.0 Console window. The window has a title bar with the Scilab logo and the text "Scilab 2025.1.0 Console". Below the title bar is a menu bar with "File", "Edit", "Control", "Applications", and "?". Under the menu bar is a toolbar with various icons for file operations, editing, and help. The main area of the console displays the output of the code: "Reflection Coefficient (K) ", -0.6666667, "Magnitude of Reflection Coefficient", 0.6666667, "Standing Wave Ratio (SWR) ", and 5.0000000. At the bottom of the console, there is a prompt "-->".

```
Scilab 2025.1.0 Console
```

```
"Reflection Coefficient (K) "  
-0.6666667  
"Magnitude of Reflection Coefficient"  
0.6666667  
"Standing Wave Ratio (SWR) "  
5.0000000  
  
-->
```